

Data Cleaning and Preparation

```
In [10]: # full code practical 3
import pandas as pd
import seaborn as sns

titanic = sns.load_dataset('titanic')
df = pd.DataFrame(titanic)
print(df.shape)
print("Missing values before cleaning")
print(df.isnull().sum())
print(df.info())
print(df.describe())
df['age'] = df['age'].fillna(df['age'].median())
df['embarked'] = df['embarked'].fillna(df['embarked'].mode()[0])
df['embark_town'] = df['embark_town'].fillna(df['embark_town'].mode()[0])
df.drop(columns=['deck'], inplace=True, errors='ignore')
df.dropna(inplace=True)
print("Missing values after cleaning")
print(df.isnull().sum())
```

```
(891, 15)
Missing values before cleaning
survived      0
pclass        0
sex           0
age          177
sibsp         0
parch         0
fare          0
embarked      2
class         0
who           0
adult_male    0
deck         688
embark_town   2
alive         0
alone         0
```

```
dtype: int64
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 891 entries, 0 to 890
```

```
Data columns (total 15 columns):
```

#	Column	Non-Null Count	Dtype
0	survived	891 non-null	int64
1	pclass	891 non-null	int64
2	sex	891 non-null	object
3	age	714 non-null	float64
4	sibsp	891 non-null	int64
5	parch	891 non-null	int64
6	fare	891 non-null	float64
7	embarked	889 non-null	object
8	class	891 non-null	category
9	who	891 non-null	object
10	adult_male	891 non-null	bool
11	deck	203 non-null	category
12	embark_town	889 non-null	object
13	alive	891 non-null	object
14	alone	891 non-null	bool

```
dtypes: bool(2), category(2), float64(2), int64(4), object(5)
```

```
memory usage: 80.7+ KB
```

```
None
```

	survived	pclass	age	sibsp	parch	fare
count	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

```
Missing values after cleaning
```

```
survived      0
pclass        0
sex           0
age           0
sibsp         0
```

```
parch      0
fare       0
embarked   0
class      0
who        0
adult_male 0
embark_town 0
alive      0
alone      0
dtype: int64
```

In detail with Explanation

```
In [11]: import pandas as pd
import seaborn as sns

# --- Step 1: Load the Titanic Dataset into a DataFrame ---
df = sns.load_dataset('titanic') # 'df' is the standard name for a DataFrame

# --- Step 2: Inspect the Dataset (Optional Steps Added) ---
print(f"Shape of dataset: {df.shape}")

print("\n--- Optional: Data Info (Data Types & Non-Null Counts) ---")
# .info() helps you see which columns are integers, floats, or objects (strings)
df.info()

print("\n--- Optional: Statistical Summary ---")
# .describe() gives you mean, std, min, max for numerical columns
print(df.describe())

print("\nMissing values before cleaning:")
print(df.isnull().sum())

# --- Step 3: Handle Missing Values ---

# 1. Fill missing 'age' with the median
# We use median because age often has outliers that skew the mean
df['age'] = df['age'].fillna(df['age'].median())

# 2. Fill missing 'embarked' and 'embark_town' with the mode
# We use [0] to get the string value from the mode Series
df['embarked'] = df['embarked'].fillna(df['embarked'].mode()[0])
df['embark_town'] = df['embark_town'].fillna(df['embark_town'].mode()[0])

# 3. Drop 'deck' column
# It has too many missing values to be useful
df.drop(columns=['deck'], inplace=True)

# 4. Drop any remaining rows with missing values
df.dropna(inplace=True)

# --- Final Check ---
print("\nMissing values after cleaning:")
print(df.isnull().sum())
```

```
print("\nFinal DataFrame Shape:")  
print(df.shape)
```

Shape of dataset: (891, 15)

--- Optional: Data Info (Data Types & Non-Null Counts) ---

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 891 entries, 0 to 890

Data columns (total 15 columns):

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dtypes: bool(2), category(2), float64(2), int64(4), object(5)

memory usage: 80.7+ KB

--- Optional: Statistical Summary ---

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alive         0
alone         0
dtype: int64
```

```
Final DataFrame Shape:
(891, 14)
```

In []: